Acminder: User Flow Analysis & Design Documentation

1. Project Brief

Problem Statement

Students and working professionals often struggle to balance multiple commitments, leading to overlapping schedules and missed deadlines. Existing calendar apps provide static views but lack proactive conflict detection and resolution strategies. There is a need for a **smart scheduling solution** that not only tracks events but also identifies conflicts and suggests actionable resolutions.

Primary Goals

The primary objective of this application is to provide a seamless experience for managing complex schedules. Key goals include:

- **Centralized Integration:** Consolidating schedules from various sources like college timetables, work shifts, and personal calendars.
- **Proactive Conflict Detection:** Automatically identifying overlapping events to prevent scheduling errors.
- **Actionable Resolution:** Offering smart suggestions to resolve conflicts, such as requesting shift changes or prioritizing tasks.
- **User-Centric Design:** Providing a clean, intuitive interface that highlights the most critical information at a glance.

Success Metrics

Metric	Description
Conflict Resolution Rate	% of detected conflicts successfully resolved using app suggestions.
Onboarding Completion	Rate at which new users successfully link their external schedules.
Daily Active Usage	Frequency of users checking the “Today” view to manage commitments.

Appendix 1: User Profiles and Personas

User Profile

Attribute	Description
Target Audience	College students with part-time jobs and busy academic schedules.
Age Range	18–25 years old.
Tech Literacy	High; comfortable with mobile apps and digital calendar integrations.
Pain Points	Overlapping shifts and classes, difficulty tracking deadlines.

User Persona: “Busy Bella”

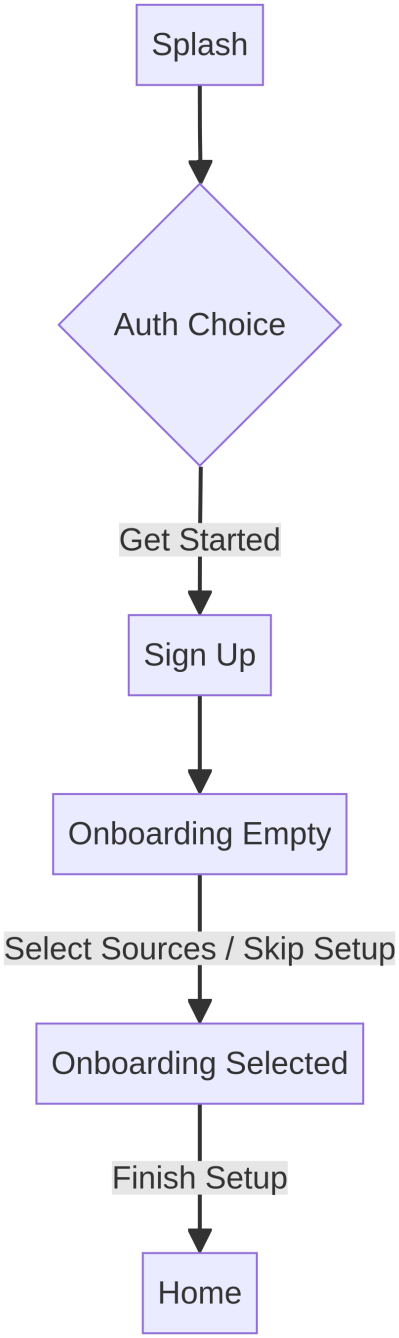
Role: Full-time Nursing Student & Part-time Barista

- **Goals:** Ensure she never misses a clinical placement while maintaining her work week.
- **Scenario:** Bella uses the Acminder to import her college timetable and sync her barista shifts. When a conflict is detected, she uses the “Request Shift Change” feature to notify her manager immediately.

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2. User Flow Diagrams

2.1. New User Onboarding Flow

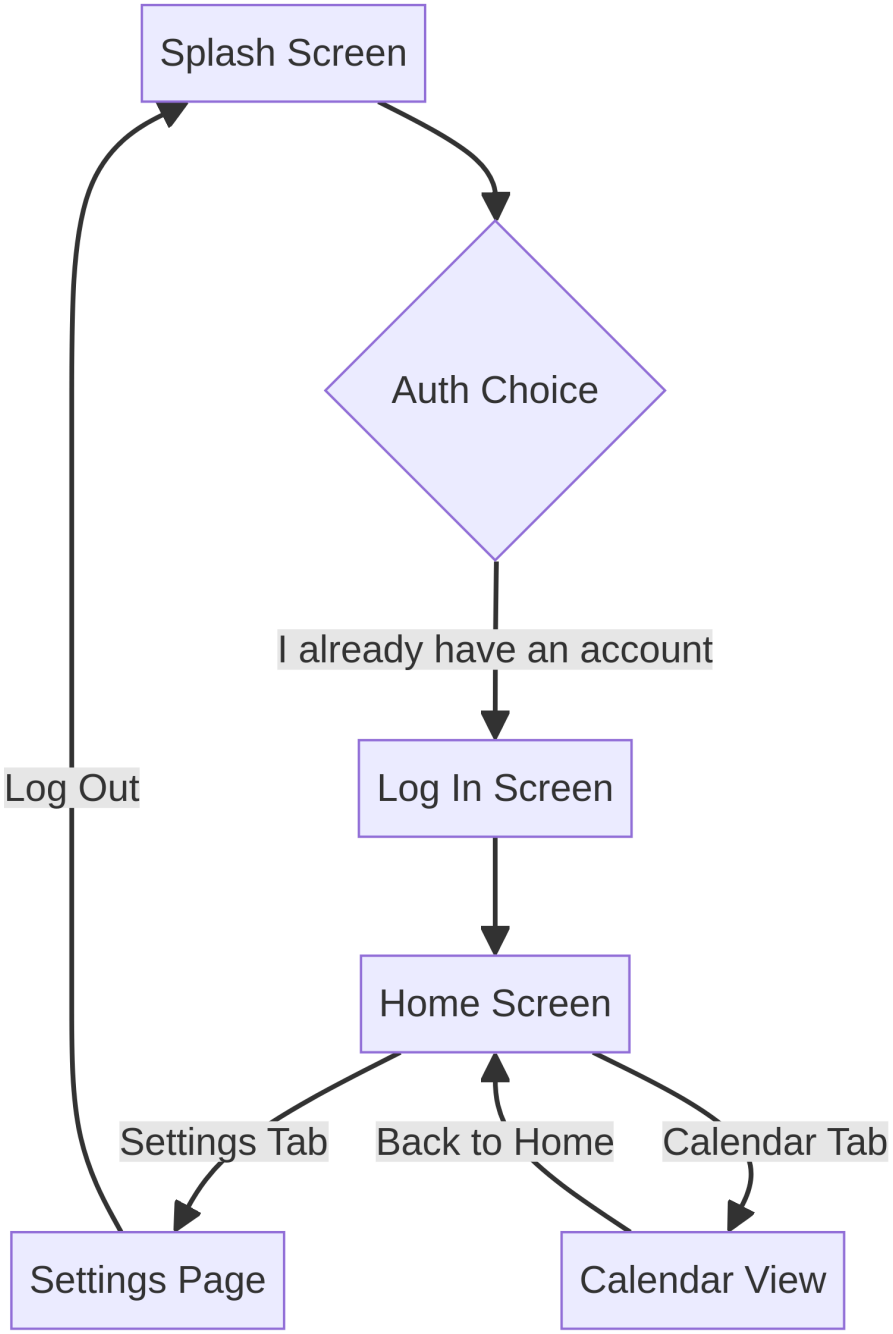


This flow outlines the journey for a new user from the initial splash screen to their first interaction with the Acminder home screen.

1. **Splash Screen:** User lands on the initial welcome screen.
2. **Authentication Choice:** User taps “Get Started” leading to a choice between “Sign Up” or “Log In”.

3. **Sign Up:** User selects “Sign Up” and provides email and password to create a new account.
4. **Onboarding - Import Schedule (Empty State):** User is prompted to connect external schedules (College Timetable, Work Shift Schedule, Google/Apple Calendar).
5. **Onboarding - Import Schedule (Selected State):** User selects one or more schedule sources or chooses to “Skip Setup”.
6. **Finish Setup:** User taps “Finish Setup” to complete the onboarding process.
7. **Home Screen:** User is directed to their personalized home screen, ready to manage their schedule.

2.2. Existing User Login Flow

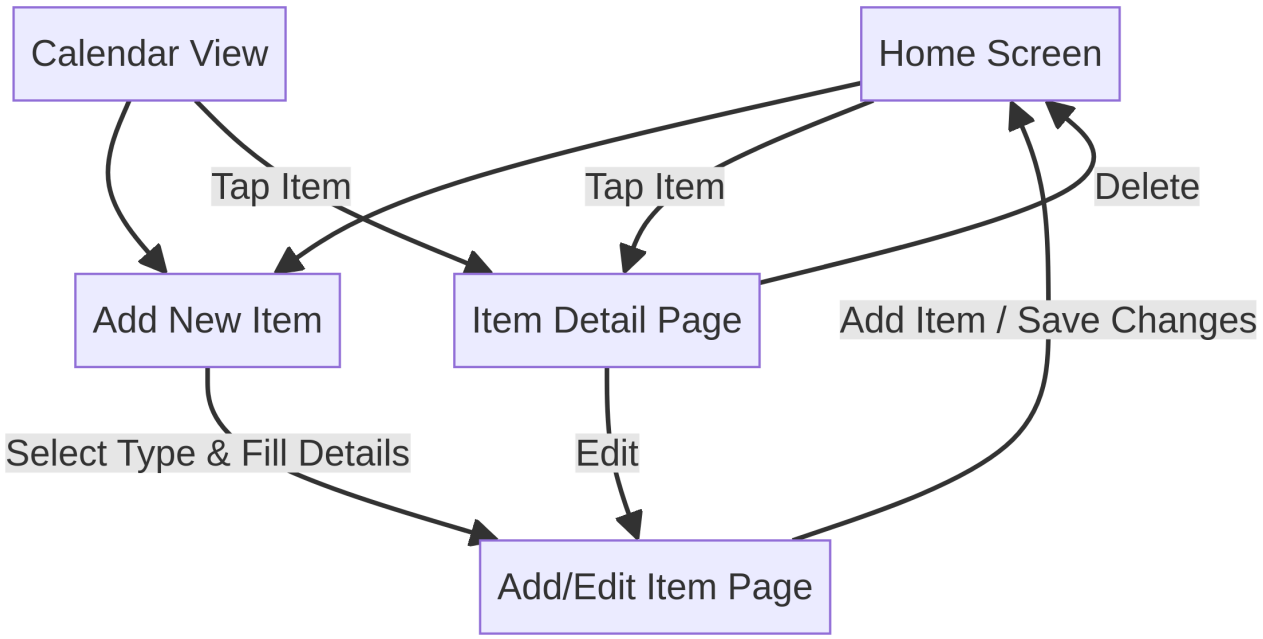


This flow describes how an existing user accesses their Acminder account and navigates to the main application features.

1. **Splash Screen:** User lands on the initial welcome screen.
2. **Authentication Choice:** User taps “I already have an account” leading to the “Log In” screen.
3. **Log In:** User provides registered email and password to log in.
4. **Home Screen:** User is directed to their personalized home screen.

5. **Navigation to Settings:** From the Home Screen, user can tap the “Settings” tab in the bottom navigation.
6. **Settings Page:** User can manage account details, preferences, and privacy settings.
7. **Log Out:** User taps “Log Out” from the Settings Page, returning to the Splash Screen.
8. **Navigation to Calendar:** From the Home Screen, user can tap the “Calendar” tab in the bottom navigation.
9. **Calendar View:** User views their schedule in a daily/weekly format.

2.3. Task Management Flow (Add, Edit, Delete)

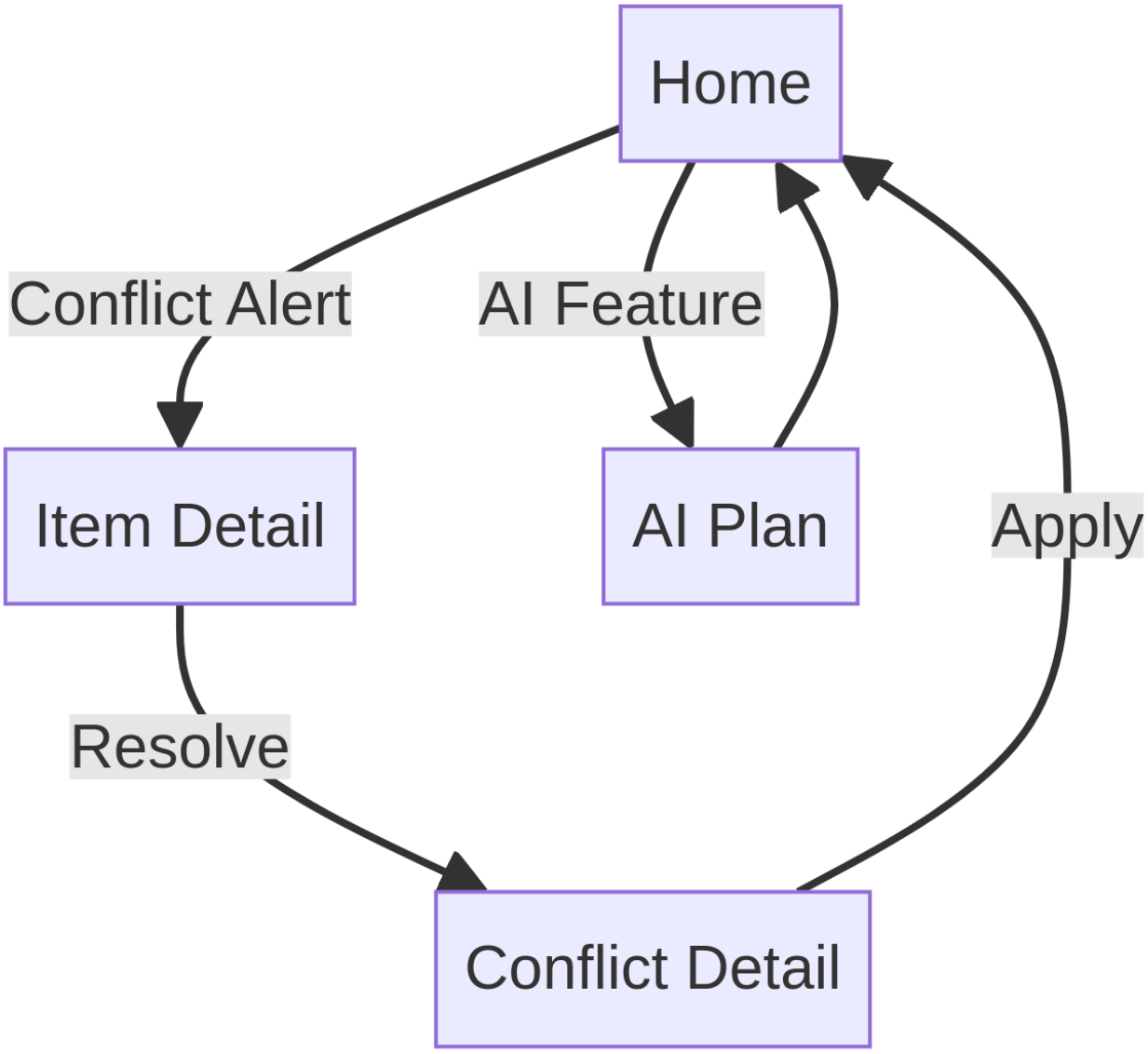


This flow details how users create, modify, and remove scheduled items within Acminder.

1. **Initiate Add Task:** From either the Home Screen or Calendar View, the user taps the Floating Action Button (FAB) or a specific time slot.
2. **Add New Item Page:** User selects the item type (Work Shift, Class, Assignment) and fills in relevant details (Title, Date, Time, Location, Role).
3. **Add Item:** User taps “Add Item” to save the new schedule entry.
4. **View Item Details:** User taps on an existing item from the Home Screen or Calendar View to see its details.

5. **Item Detail Page:** Displays comprehensive information about the selected item.
6. **Edit Item:** From the Item Detail Page, user taps “Edit” to modify the item.
7. **Edit Item Page:** User updates the item details.
8. **Save Changes:** User taps “Save Changes” to apply modifications.
9. **Delete Item:** From the Item Detail Page, user taps “Delete” (often followed by a confirmation step).

2.4. Conflict Resolution & AI Weekly Planning Flow



This flow highlights Acminder’s core intelligent features: detecting and resolving schedule conflicts, and conceptual AI-driven weekly planning.

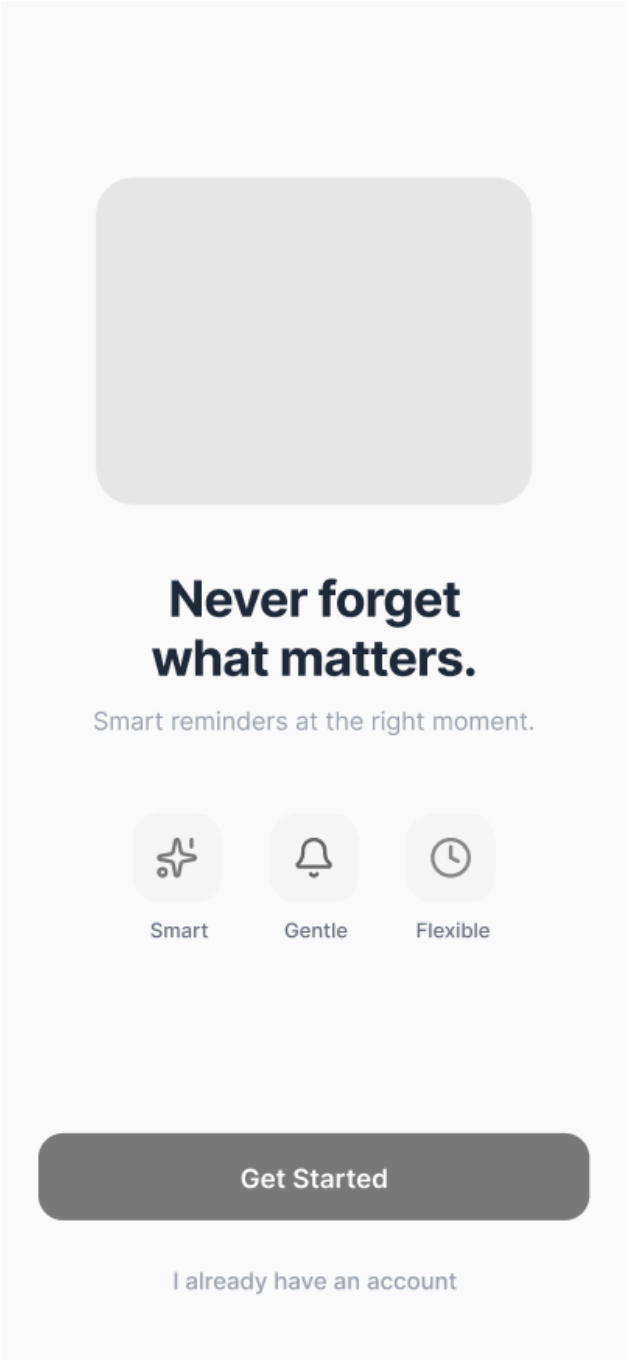
1. **Conflict Detection:** Acminder automatically identifies overlapping schedule items, displaying a “Conflict Detected” alert on the Home Screen.
2. **View Conflict:** User taps the “Conflict Detected” alert or a conflict-flagged item on the Home Screen, leading to the Item Detail Page (Conflict).
3. **Resolve Conflict:** From the Item Detail Page, user taps “Resolve Conflict” to proceed.
4. **Conflict Detail Page:** Displays conflicting items and offers resolution suggestions (e.g., “Request Shift Change”, “Finish Assignment Asap”).
5. **Apply Resolution:** User selects a suggestion and taps “Apply Resolution” to enact the chosen solution.
6. **Return to Home Screen:** The schedule is updated, and the user returns to the Home Screen.
7. **Generate Weekly Plan by AI (Conceptual):** From the Home Screen (or a dedicated feature), the user initiates the AI planning feature to generate an optimized weekly schedule. (This is a conceptual future feature based on the app’s intelligent premise.)
8. **Updated Home Screen:** The Home Screen reflects the AI-generated plan.

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3. Annotated Wireframes v1

This section provides detailed annotations for the major screens, focusing on interaction, content hierarchy, and accessibility, with the wireframe images embedded directly.

1. Splash Screen



Screen Name: Splash Screen

Description: The initial entry point for the application, designed to introduce the app's core value proposition and guide new or returning users to the next step.

Key Elements & Annotations:

- **Visual Placeholder:** A large, central grey box serves as a placeholder for potential future branding or illustrative graphics, emphasizing a clean and

minimalist aesthetic.

- **Headline (`Never forget what matters.`):** Clearly communicates the app's primary benefit, focusing on memory and importance.
- **Sub-headline (`Smart reminders at the right moment.`):** Elaborates on the headline, highlighting the intelligent and timely nature of the reminders.
- **Feature Icons (`Smart`, `Gentle`, `Flexible`):** Three distinct icons with accompanying text visually represent the key attributes of the reminder system, offering a quick overview of functionality.
- **Primary Call-to-Action (`Get Started` button):** A prominent, full-width button encourages new users to begin their journey with the application.
- **Secondary Call-to-Action (`I already have an account` link):** A less prominent text link provides an option for existing users to log in, maintaining a clear hierarchy of actions.

Interaction & Behavior:

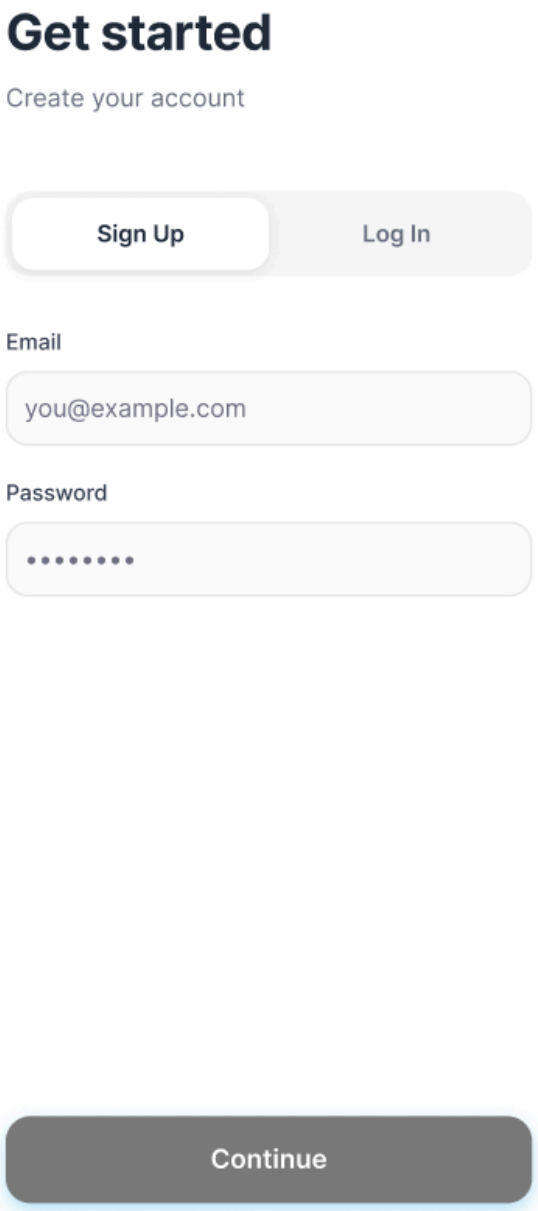
- Tapping the `Get Started` button navigates the user to the Sign Up screen.
- Tapping the `I already have an account` link navigates the user to the Log In screen.

Accessibility Considerations:

- The contrast ratio between the text and background is high, ensuring readability. The primary call-to-action is clearly distinguishable.

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2. Sign Up Screen



Screen Name: Sign Up Screen

Description: Facilitates the creation of a new user account, providing a straightforward process for entering essential credentials.

Key Elements & Annotations:

- **Headline (`Get started` / `Create your account`):** A clear and concise title indicating the purpose of the screen.

- **Authentication Toggle (`Sign Up` / `Log In`):** A segmented control allowing users to switch between signing up and logging in without leaving the page. `Sign Up` is highlighted as the active state.
- **Email Input Field (`you@example.com` placeholder):** A standard text input for the user's email address, with a descriptive placeholder.
- **Password Input Field (masked):** A secure text input for the password, with characters masked for privacy.
- **Primary Call-to-Action (`Continue` button):** A full-width button to proceed with the account creation. This button should be disabled until both email and password fields meet validation criteria.

Interaction & Behavior:

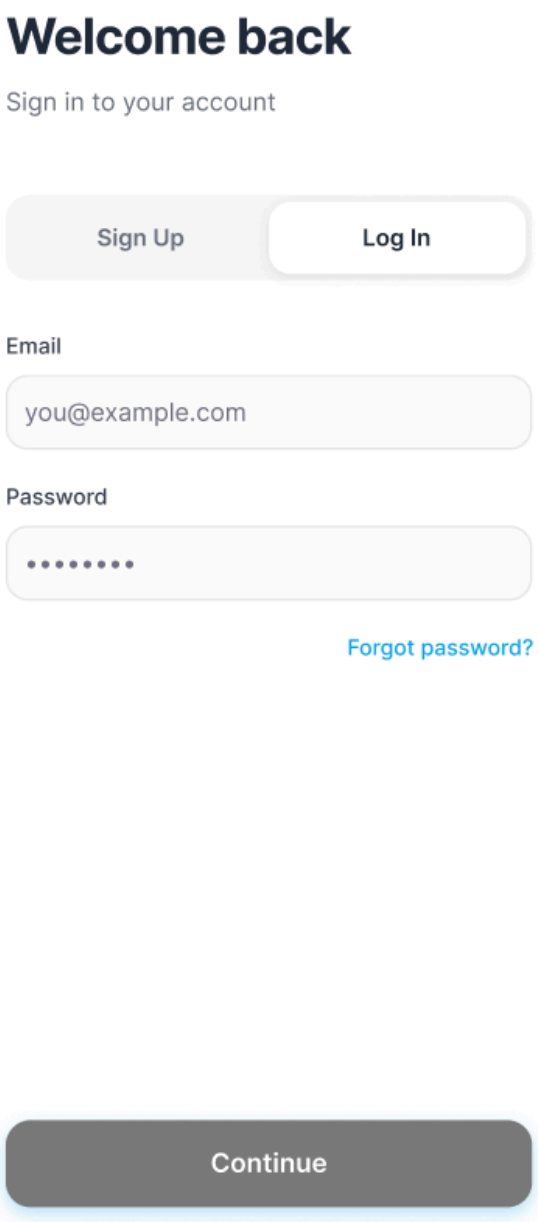
- Users enter their email and a new password.
- The `Continue` button becomes active upon valid input.
- Tapping `Log In` on the toggle switches to the Log In screen.

Accessibility Considerations:

- Input fields have clear labels and sufficient touch targets. Password masking is standard, but a toggle to reveal the password (not shown here) would enhance usability and accessibility.

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3. Log In Screen



Screen Name: Log In Screen

Description: Allows existing users to access their accounts by providing their registered credentials.

Key Elements & Annotations:

- **Headline (`Welcome back` / `Sign in to your account`):** A welcoming title that clearly states the screen's purpose.

- **Authentication Toggle (`Sign Up` / `Log In`):** Similar to the Sign Up screen, this toggle allows switching between authentication methods. `Log In` is highlighted as the active state.
- **Email Input Field (`you@example.com` placeholder):** For entering the registered email address.
- **Password Input Field (masked):** For entering the account password.
- **Forgot Password Link (`Forgot password?`):** A clickable link to initiate the password recovery process, crucial for user support.
- **Primary Call-to-Action (`Continue` button):** To submit the login credentials. This button should be disabled until both email and password fields have valid input.

Interaction & Behavior:

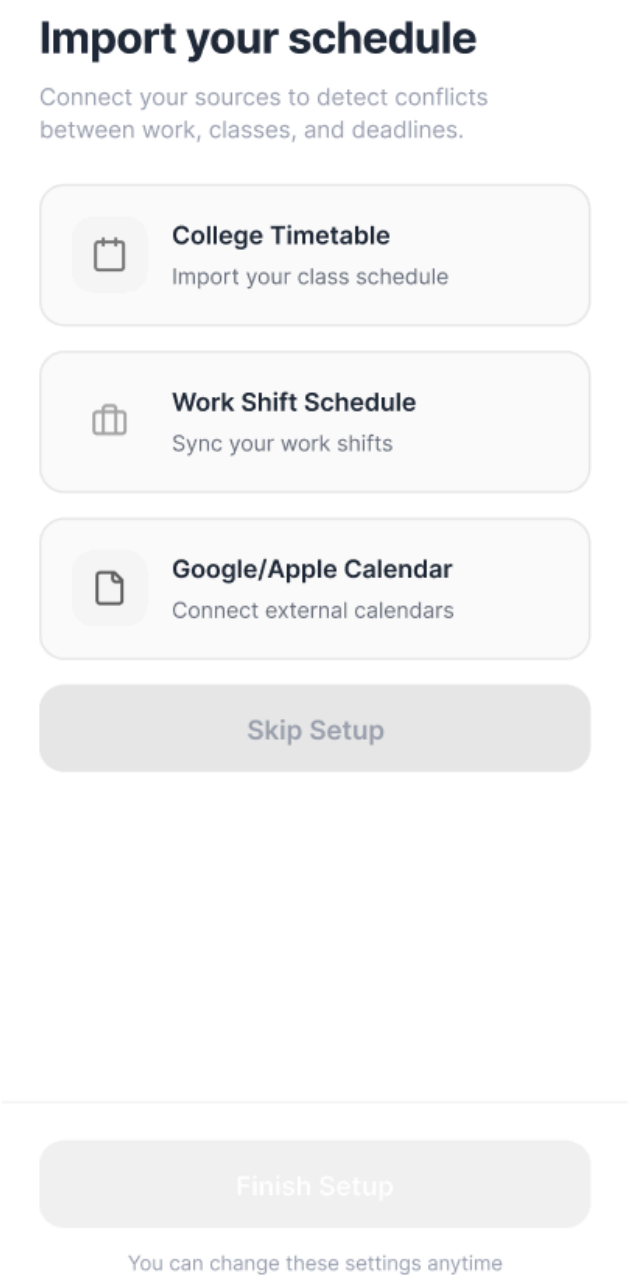
- Users enter their registered email and password.
- The `Continue` button becomes active upon valid input.
- Tapping `Sign Up` on the toggle switches to the Sign Up screen.
- Tapping `Forgot password?` navigates to a password recovery flow (not shown).

Accessibility Considerations:

- Clear labels and sufficient touch targets for input fields. The Forgot password? link is clearly visible and clickable.

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4. Onboarding - Import Schedule (Empty State)



Screen Name: Onboarding - Import Schedule (Empty State)

Description: This screen guides new users through the process of integrating their existing schedules to enable conflict detection, presented before any sources are selected.

Key Elements & Annotations:

- **Headline (`Import your schedule`):** Clearly states the purpose of this onboarding step.
- **Description (`Connect your sources to detect conflicts between work, classes, and deadlines.`):** Explains the benefit of importing schedules, motivating user action.
- **Integration Options (`College Timetable`, `Work Shift Schedule`, `Google/Apple Calendar`):** Three distinct cards representing different schedule sources. Each card includes an icon, title, and a brief description.
- **Secondary Action (`Skip Setup` button):** Allows users to bypass schedule integration if they prefer to add items manually or return later.
- **Primary Call-to-Action (`Finish Setup` button - disabled):** A prominent button to complete the setup. It is initially disabled, indicating that a selection is required or the setup must be skipped.

Interaction & Behavior:

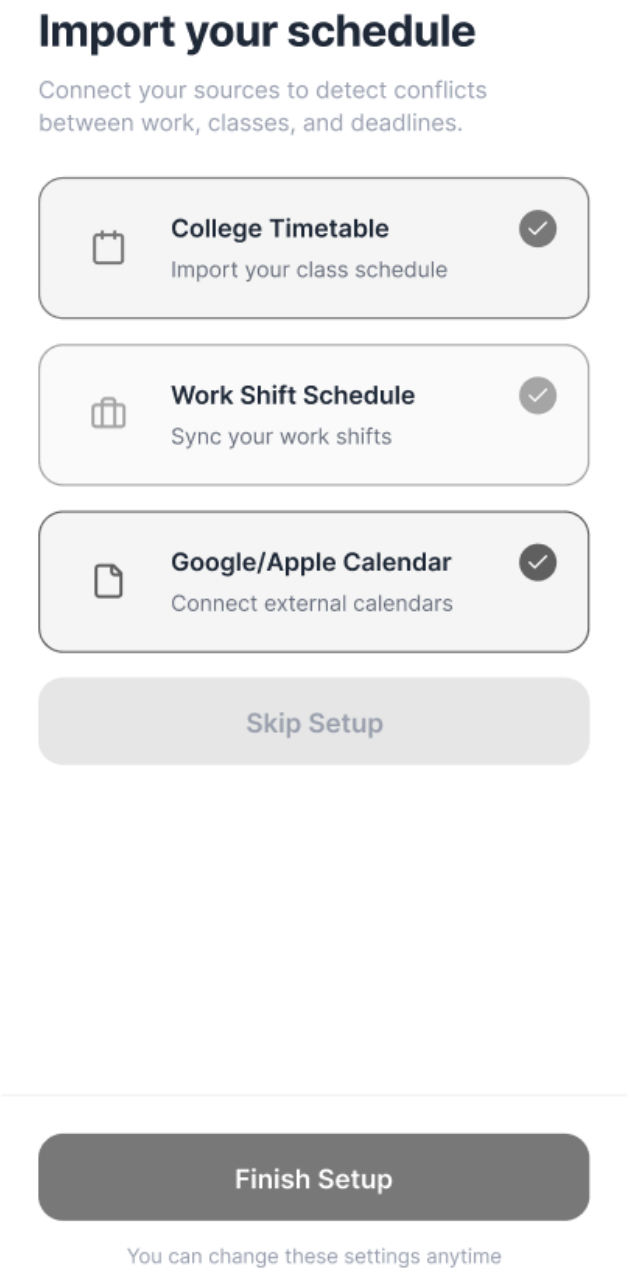
- Tapping any of the integration option cards will mark it as selected (as seen in the next wireframe).
- The `Finish Setup` button remains disabled until at least one integration source is selected or the `Skip Setup` button is tapped.
- Tapping `Skip Setup` proceeds to the Home screen without importing schedules.

Accessibility Considerations:

- Large, clearly defined cards for integration options ensure easy tapping. The disabled state of the `Finish Setup` button provides visual feedback.

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5. Onboarding - Import Schedule (Selected State)



Screen Name: Onboarding - Import Schedule (Selected State)

Description: This screen shows the state after a user has selected one or more schedule integration options, enabling the completion of the setup.

Key Elements & Annotations:

- **Integration Options (with checkmarks):** The same three integration cards, but now with a checkmark icon indicating that they have been selected or

successfully connected.

- **Primary Call-to-Action (`Finish Setup` button - enabled):** The button is now active, allowing the user to finalize the onboarding process.

Interaction & Behavior:

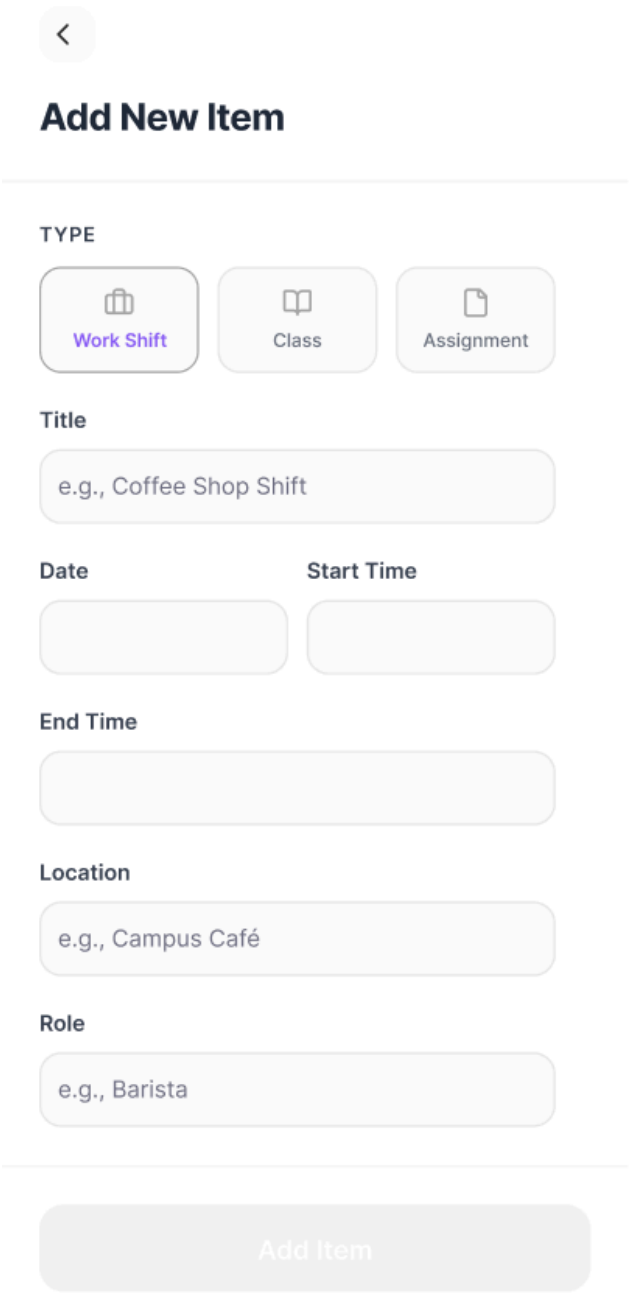
- Tapping the `Finish Setup` button navigates the user to the Home screen, presumably with their imported schedules now integrated.
- Users can deselect options if they change their mind before finishing setup.

Accessibility Considerations:

- The checkmark provides clear visual confirmation of selection. The enabled state of the `Finish Setup` button is distinct from its disabled state.

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6. Add New Item - Work Shift



Screen Name: Add New Item - Work Shift

Description: A form for users to manually add a new work shift to their schedule.

Key Elements & Annotations:

- **Back Button:** Allows the user to return to the previous screen (e.g., Calendar or Home).
- **Headline (`Add New Item`):** Clearly indicates the screen's function.

- **Item Type Selector (`Work Shift`, `Class`, `Assignment`):** A segmented control to choose the type of item being added. `Work Shift` is currently selected, which dictates the visible form fields.
- **Title Input Field (`e.g., Coffee Shop Shift`):** Text input for the name of the work shift.
- **Date Input Field:** For selecting the date of the shift.
- **Start Time Input Field:** For selecting the beginning time of the shift.
- **End Time Input Field:** For selecting the end time of the shift.
- **Location Input Field (`e.g., Campus Café`):** Text input for the location of the work shift.
- **Role Input Field (`e.g., Barista`):** Text input for the user's role during the shift.
- **Primary Call-to-Action (`Add Item` button - disabled):** This button is initially disabled and should only become active once all mandatory fields are filled with valid data.

Interaction & Behavior:

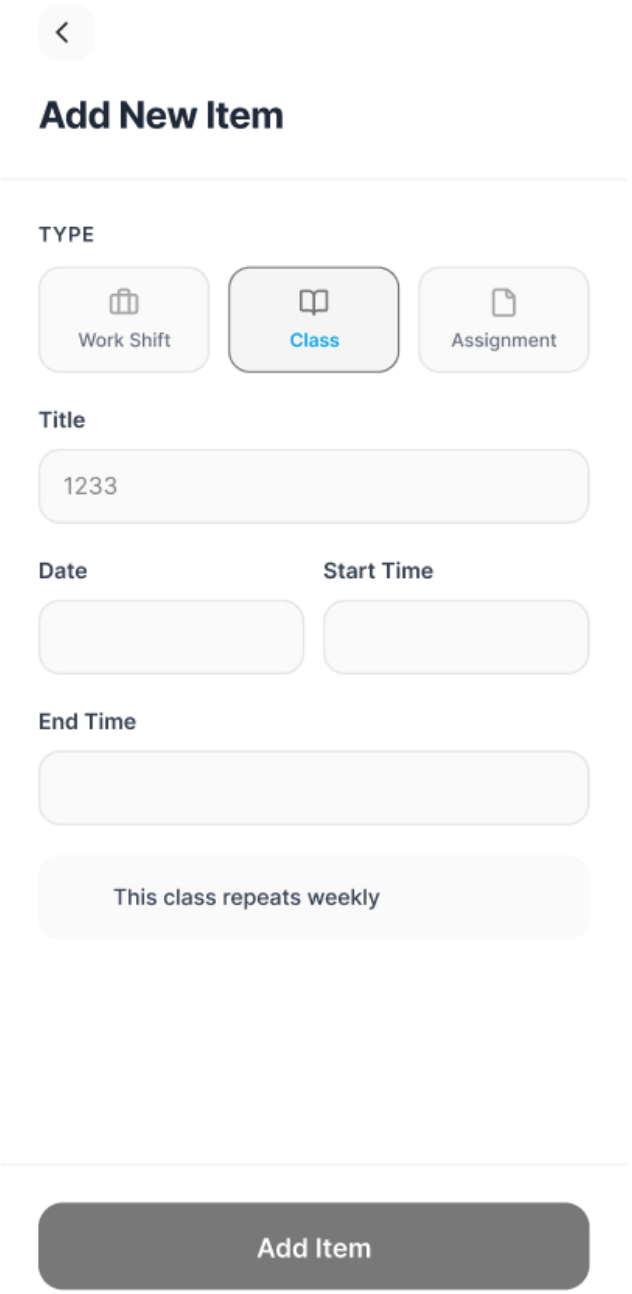
- Selecting a different item type (e.g., `Class` or `Assignment`) would dynamically change the visible input fields to be relevant to that item type.
- Date and time fields would likely trigger native date/time pickers for ease of input.
- The `Add Item` button activates upon successful validation of all required fields.

Accessibility Considerations:

- Clear labels and placeholder text guide user input. The dynamic nature of the form based on item type helps reduce cognitive load.

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7. Add New Item - Class



Screen Name: Add New Item - Class

Description: A form for users to manually add a new class to their schedule, demonstrating how the form adapts to different item types.

Key Elements & Annotations:

- **Item Type Selector (`Class` selected):** `Class` is now selected, and the form fields have adapted accordingly.

- **Title Input Field:** For the class name.
- **Date Input Field:** For the class date.
- **Start Time Input Field:** For the class start time.
- **End Time Input Field:** For the class end time.
- **Repetition Note (`This class repeats weekly`):** A text note indicating a recurring nature, suggesting an underlying logic or a future option for setting recurrence.
- **Primary Call-to-Action (`Add Item` button - enabled):** The button is enabled, implying that the necessary fields for a class item have been filled.

Interaction & Behavior:

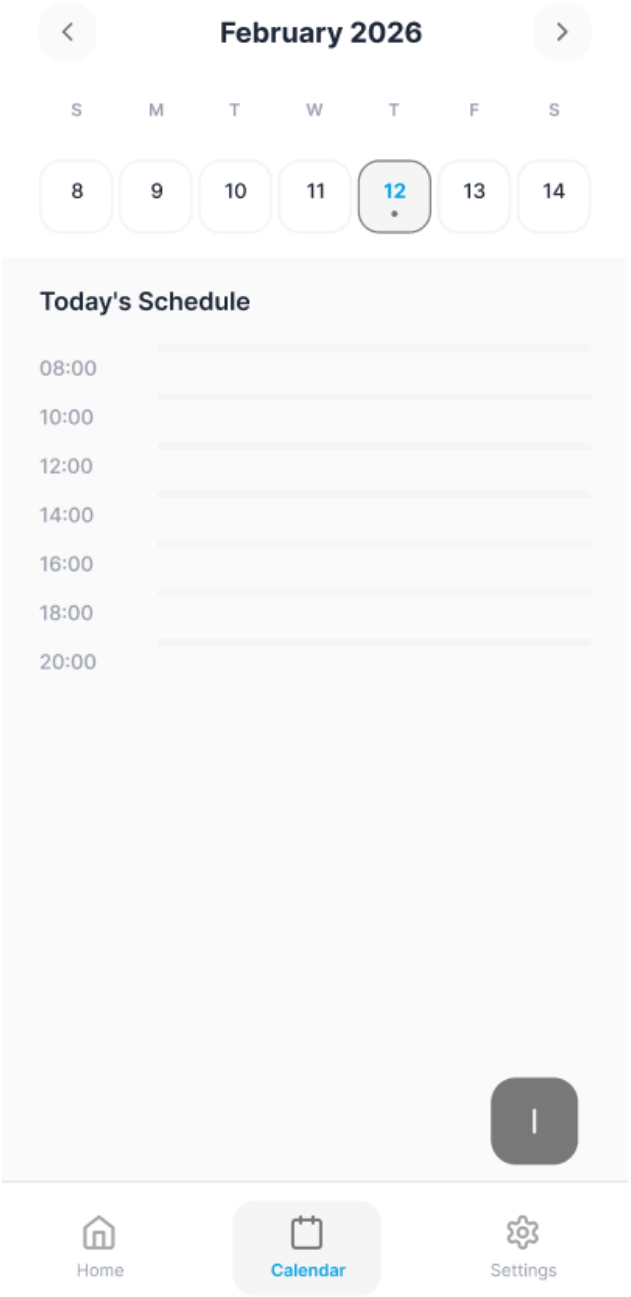
- The form fields are streamlined for class-specific information, omitting 'Location' and 'Role' which were present for 'Work Shift'.
- The repetition note suggests an automatic detection or a pre-set option for recurring events.

Accessibility Considerations:

- The form's adaptability reduces clutter and focuses user attention on relevant inputs for the selected item type.

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8. Calendar View



Screen Name: Calendar View

Description: Provides a daily or weekly overview of the user's schedule, allowing for navigation and quick addition of new items.

Key Elements & Annotations:

- **Navigation Header (`February 2026`):** Displays the current month and year, with left/right arrows for navigating to previous/next months.

- **Weekly Date Picker:** A horizontal scrollable list of days, with the current day (`12`) highlighted. A small dot under the `12` might indicate events on that day.
- **Today's Schedule (Time Slots):** A vertical list of time slots (e.g., `08:00`, `10:00`) indicating the daily schedule. This area would populate with scheduled items.
- **Floating Action Button (FAB):** A prominent circular button, typically used for the primary action of the screen, which in this context is likely to `Add New Item`.
- **Bottom Navigation Bar (`Home`, `Calendar`, `Settings`):** Provides quick access to the main sections of the application. `Calendar` is highlighted as the active screen.

Interaction & Behavior:

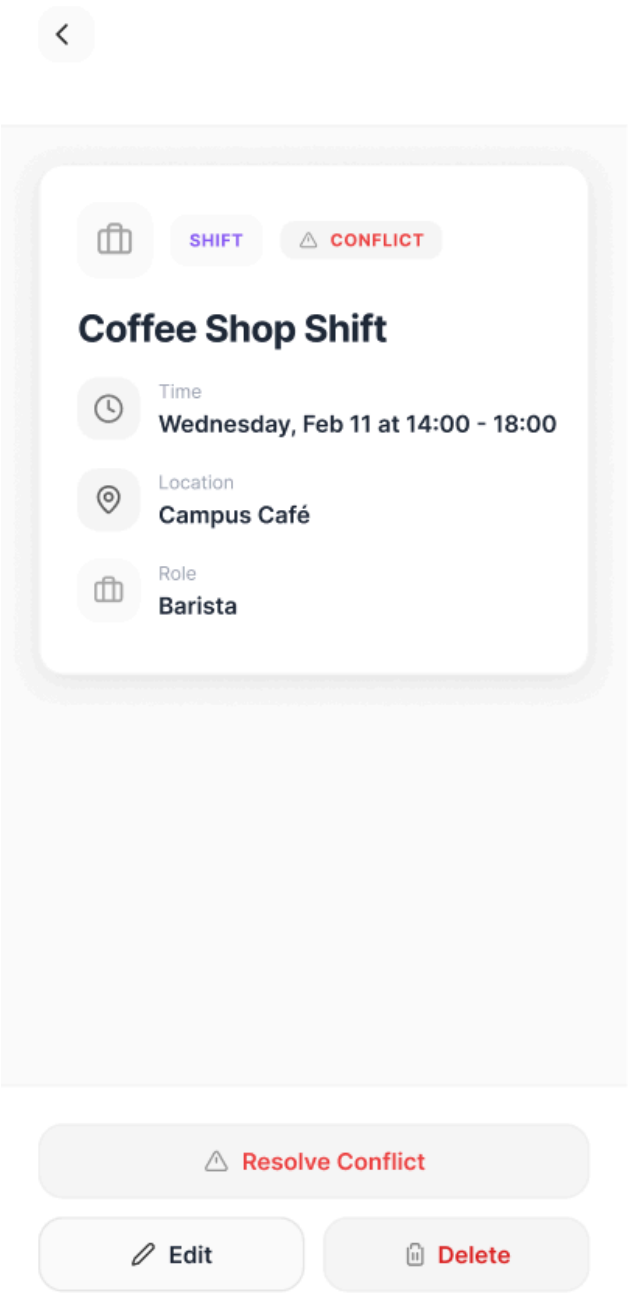
- Tapping on a different day in the weekly date picker would update the `Today's Schedule` section to show events for that day.
- Tapping the FAB would navigate to the `Add New Item` screen.
- Swiping left/right on the weekly date picker or using the arrows in the header would change the visible week/month.

Accessibility Considerations:

- The active state of the current day and the active tab in the bottom navigation are clearly indicated. The FAB is large and easily tappable.

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9. Item Detail Page



Screen Name: Item Detail Page

Description: Displays comprehensive information about a specific scheduled item, particularly when a conflict is detected.

Key Elements & Annotations:

- **Back Button:** Returns the user to the previous screen.
- **Item Card:** A central card containing all details of the scheduled item.

    - **Type Tags (`SHIFT`, `CONFLICT`):** Labels indicating the nature of the item and the presence of a conflict. The `CONFLICT` tag is highlighted in red, drawing immediate attention.
    - **Title (`Coffee Shop Shift`):** The name of the scheduled item.
    - **Details (`Time`, `Location`, `Role`):** Specific attributes of the item, each with an icon and its corresponding value.
- **Primary Call-to-Action (`Resolve Conflict` button):** A prominent button, outlined in red, prompting the user to address the detected conflict.
- **Secondary Actions (`Edit`, `Delete` buttons):** Allow users to modify or remove the scheduled item.

Interaction & Behavior:

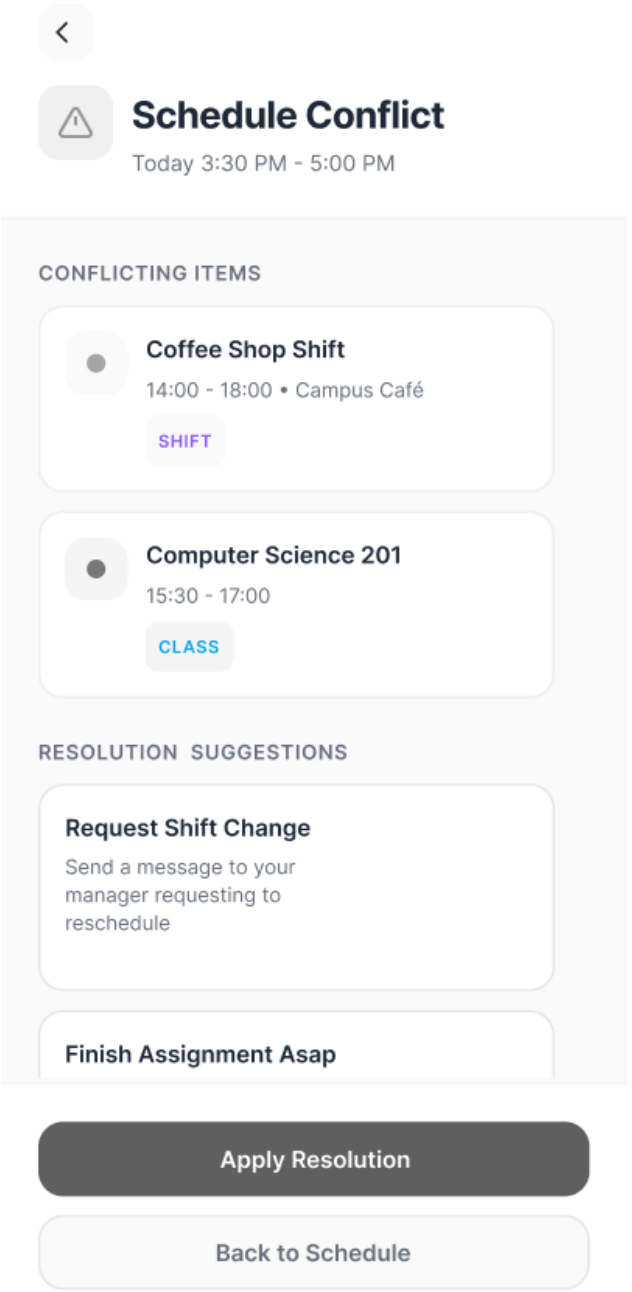
- Tapping `Resolve Conflict` navigates to the `Conflict Detail Page`.
- Tapping `Edit` navigates to the `Edit Item Page`.
- Tapping `Delete` would likely trigger a confirmation dialog before removal.

Accessibility Considerations:

- The use of both a red color and a warning icon for `CONFLICT` ensures visibility for color-blind users. Clear labeling of actions (`Edit`, `Delete`) is important.

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10. Conflict Detail Page



Screen Name: Conflict Detail Page

Description: Provides a detailed view of a schedule conflict, presenting the conflicting items and offering suggestions for resolution.

Key Elements & Annotations:

- **Back Button:** Returns to the previous screen.
- **Headline (`Schedule Conflict`):** Clearly identifies the screen's purpose.

- **Conflict Timeframe:** Displays the duration of the conflict (e.g., `Today 3:30 PM - 5:00 PM`).
- **Conflicting Items Section:** Lists the individual items that are overlapping.
    - **Item Details (`Coffee Shop Shift`, `Computer Science 201`):** Each conflicting item is presented with its name, time, and type tag (`SHIFT`, `CLASS`).
- **Resolution Suggestions Section:** Offers actionable options to resolve the conflict.
    - **Suggestions (`Request Shift Change`, `Finish Assignment Asap`):** Each suggestion is presented as a card with a title and a brief description of the action.
- **Primary Call-to-Action (`Apply Resolution` button):** A prominent button to enact the chosen resolution.
- **Secondary Action (`Back to Schedule` button):** Allows the user to return to their schedule without applying a resolution.

Interaction & Behavior:

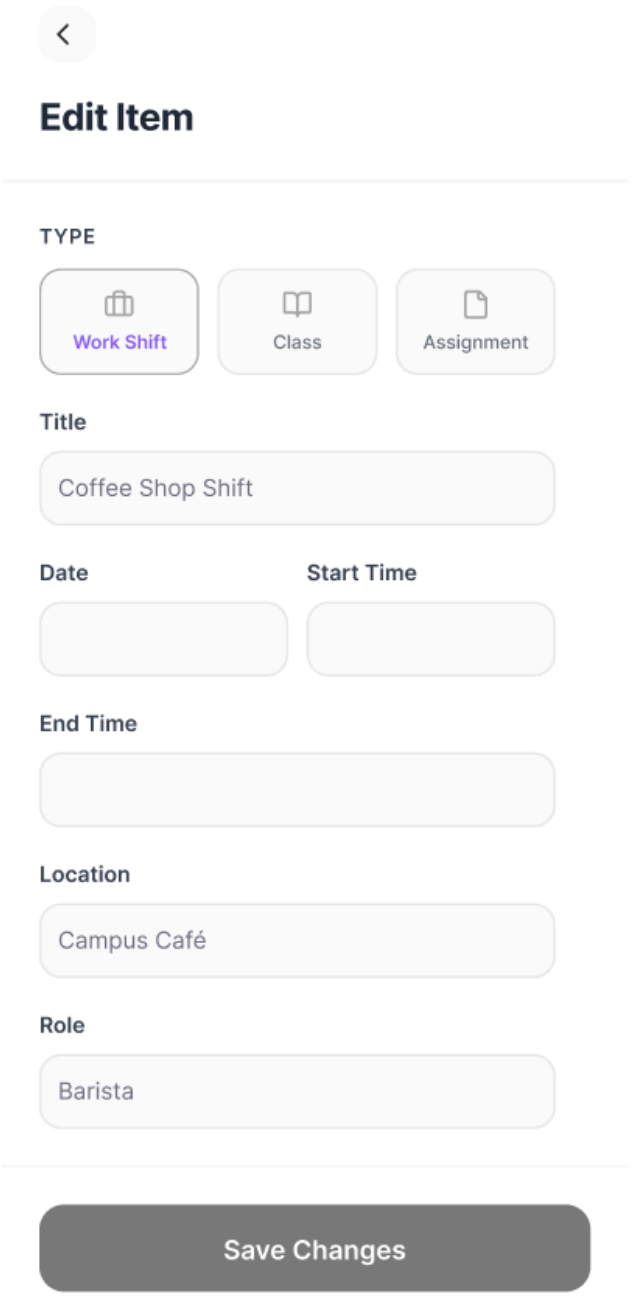
- Users can tap on a suggestion to select it. The `Apply Resolution` button would then become active.
- Tapping `Apply Resolution` would initiate the chosen action (e.g., sending a message for a shift change).

Accessibility Considerations:

- Clear separation of conflicting items and resolution suggestions improves readability. The suggestions are presented as distinct, tappable cards.

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11. Edit Item Page



Screen Name: Edit Item Page

Description: Allows users to modify the details of an existing scheduled item.

Key Elements & Annotations:

- **Back Button:** Returns to the previous screen.
- **Headline (`Edit Item`):** Clearly indicates the screen's function.

- **Item Type Selector (`Work Shift`, `Class`, `Assignment`):** Similar to the `Add New Item` screen, this allows changing the item type, which would dynamically update the form fields. `Work Shift` is selected, and fields are pre-filled with existing data.
- **Pre-filled Input Fields:** All relevant fields (Title, Date, Start Time, End Time, Location, Role) are pre-populated with the current item's data, making it easy for users to make changes.
- **Primary Call-to-Action (`Save Changes` button):** A prominent button to commit the modifications. This button should be enabled only when changes have been made and validated.

Interaction & Behavior:

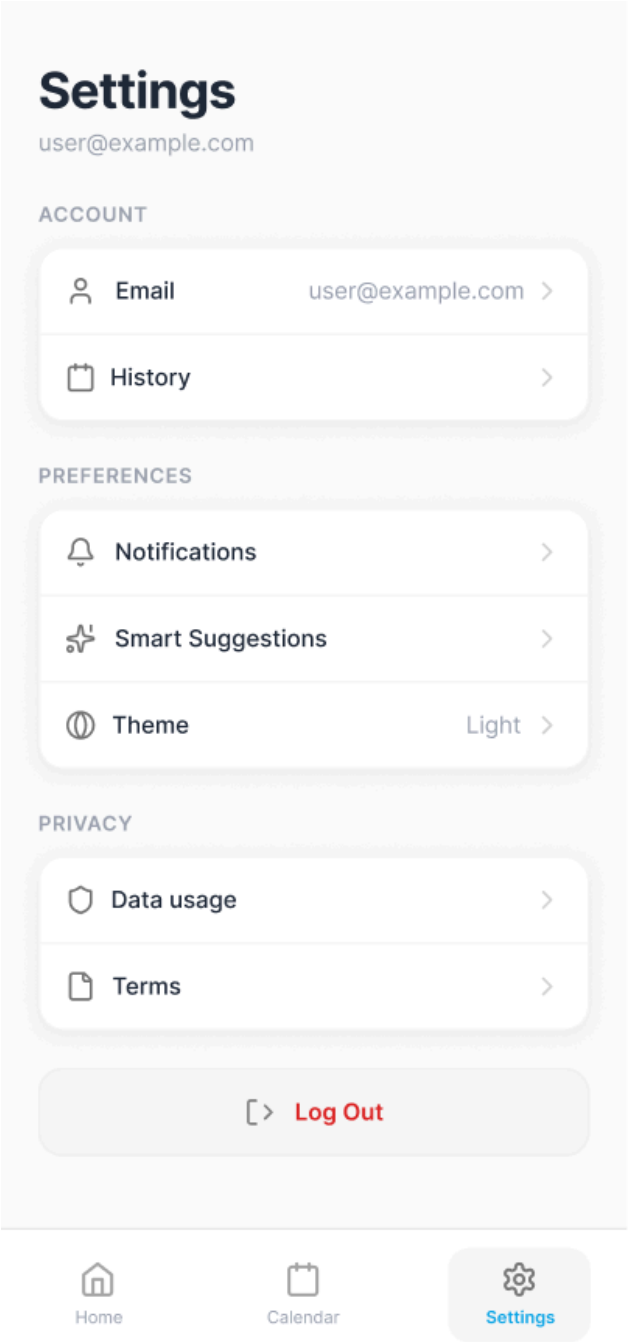
- Users can modify any of the pre-filled fields.
- The `Save Changes` button activates when changes are detected and valid.
- Changing the item type would refresh the form with fields relevant to the new type, potentially requiring re-entry of some data.

Accessibility Considerations:

- Pre-filled fields reduce the effort required for editing. Clear labels and sufficient touch targets are maintained.

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12. Settings Page



Screen Name: Settings Page

Description: Provides access to account management, application preferences, and privacy settings.

Key Elements & Annotations:

- **Headline (`Settings`):** Clearly identifies the screen.
- **User Email Display:** Shows the logged-in user's email address at the top.

- **Account Section:**
  - **Email:** Displays the user's email, potentially leading to an email change screen.
  - **History:** Likely navigates to a log of past activities or changes.
- **Preferences Section:**
  - **Notifications:** Manages app notification settings.
  - **Smart Suggestions:** Controls the behavior of AI-driven suggestions.
  - **Theme (`Light`):** Allows users to switch between different visual themes (e.g., Light/Dark mode).
- **Privacy Section:**
  - **Data usage:** Provides information or controls related to how user data is utilized.
  - **Terms:** Displays the application's terms of service.
- **Log Out Button:** A prominent red button to securely log out of the application.
- **Bottom Navigation Bar (`Home`, `Calendar`, `Settings`):** `Settings` is highlighted as the active screen.

Interaction & Behavior:

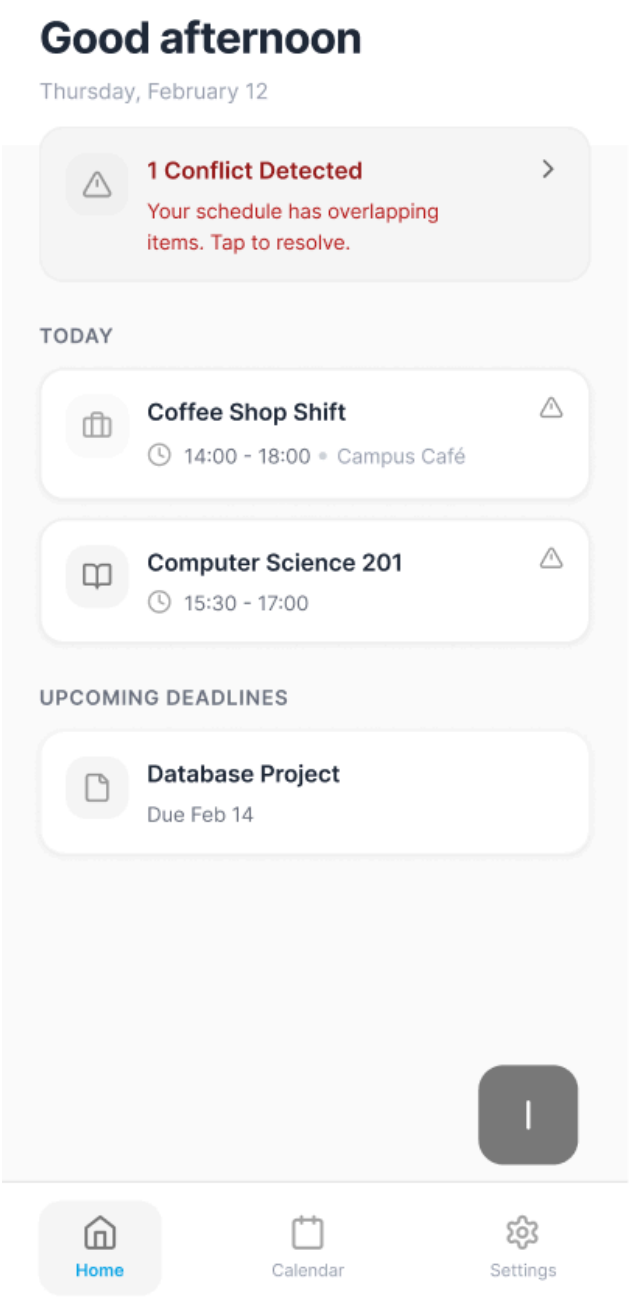
- Tapping on any of the listed options (e.g., `Notifications`, `Theme`) would navigate to a sub-screen for managing that specific setting.
- Tapping `Log Out` would sign the user out and return them to the Splash Screen.

Accessibility Considerations:

- Grouped settings into logical sections for easier navigation. The `Log Out` button is clearly distinguished by its color and placement.

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13. Home Screen / User Profile Screen



Screen Name: Home Screen / User Profile Screen

Description: The main dashboard of the application, providing an overview of the day's schedule, upcoming deadlines, and immediate conflict alerts.

Key Elements & Annotations:

- **Greeting (`Good afternoon`):** A personalized greeting based on the time of day.
- **Date Display (`Thursday, February 12`):** Shows the current day and date.

- **Conflict Alert (`1 Conflict Detected`):** A prominent red banner at the top, indicating an overlapping schedule. It includes a warning icon and a call to action (`Tap to resolve`).
- **Today's Section:** Lists scheduled items for the current day.
  - **Scheduled Items (`Coffee Shop Shift`, `Computer Science 201`):** Each item displays its name, time, and location/type. Both items have a warning icon, indicating they are part of the detected conflict.
- **Upcoming Deadlines Section:** Lists important deadlines.
  - **Deadline Item (`Database Project`):** Displays the project name and due date.
- **Floating Action Button (FAB):** A circular button for quickly adding new items.
- **Bottom Navigation Bar (`Home`, `Calendar`, `Settings`):** `Home` is highlighted as the active screen.

Interaction & Behavior:

- Tapping the `Conflict Alert` banner navigates to the `Conflict Detail Page`.
- Tapping on individual scheduled items (e.g., `Coffee Shop Shift`) would navigate to their respective `Item Detail Page`.
- Tapping the FAB would navigate to the `Add New Item` screen.

Accessibility Considerations:

- The conflict alert uses both color and an icon for accessibility. Clear headings and visual separation of sections improve readability and scannability.

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4. Reflection & Change Control

Change Control (v1)

- **Version:** 1.0
- **Status:** Initial Submission.
- **Note:** Focused on establishing the core MVP flows: Onboarding, Conflict Resolution, and Item Management.

Reflection Report

The Acminder was designed with a **utility-first** approach. We prioritized a clean, “light” theme to promote a sense of calm for users managing busy schedules. Key decisions included placing the “Conflict Detected” alert at the top of the Home screen for immediate visibility and providing functional onboarding that imports data right away. By offering specific resolution suggestions rather than just identifying problems, we aim to provide a more proactive and helpful user experience. Next steps include high-fidelity prototyping and user testing on the resolution flow to ensure intuitive interactions.